

# Abdul Rahman Kreidieh

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| CONTACT    | akreidieh@gmail.com — <a href="https://github.com/aboudykreidieh">aboudykreidieh.github.io</a> — <a href="https://linkedin.com/in/abdurahmankreidieh">linkedin.com/in/abdurahmankreidieh</a>                                                                                                                                                                                                                                                                                    |                                       |
| EDUCATION  | <b>University of California, Berkeley</b>                                                                                                                                                                                                                                                                                                                                                                                                                                       | <i>August 2016 - December 2022</i>    |
|            | <ul style="list-style-type: none"><li>• MSc/Ph.D. in Civil and Environmental Engineering</li><li>• Advised by Alexandre M. Bayen</li><li>• Areas of research: Reinforcement learning, autonomous vehicles, traffic control</li></ul>                                                                                                                                                                                                                                            |                                       |
|            | <b>American University of Beirut</b>                                                                                                                                                                                                                                                                                                                                                                                                                                            | <i>September 2012 - May 2016</i>      |
|            | <ul style="list-style-type: none"><li>• Bachelors in Mechanical Engineering</li></ul>                                                                                                                                                                                                                                                                                                                                                                                           |                                       |
| EXPERIENCE | <b>Software Engineer, YouTube</b>                                                                                                                                                                                                                                                                                                                                                                                                                                               | <i>April 2025 - present</i>           |
|            | <ul style="list-style-type: none"><li>• Implementing techniques and algorithms to improve the recommendation quality of YouTube Autoplay.</li><li>• Adapting and advancing the state-of-the-art in reinforcement learning for recommender systems.</li></ul>                                                                                                                                                                                                                    |                                       |
|            | <b>Research Scientist, Google Research</b>                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>March 2023 - April 2025</i>        |
|            | <ul style="list-style-type: none"><li>• Coordinated with teams in Research, Maps, and Ads on a variety of geospatial reasoning problems.</li><li>• Conducted a statistical study on the effect of New York City's congestion pricing policy.</li><li>• Constructed ML models to integrate geospatial and mobility data into ads-related tasks.</li><li>• Assisted in the largest study to date demonstrating that intelligent routing strategies can improve traffic.</li></ul> |                                       |
|            | <b>Graduate Student Researcher, UC Berkeley</b>                                                                                                                                                                                                                                                                                                                                                                                                                                 | <i>May 2018 - December 2022</i>       |
|            | <ul style="list-style-type: none"><li>• Designed a platform to run reinforcement learning experiments in simulations of mixed autonomy traffic.</li><li>• Developed a technique to improve the training stability of hierarchical reinforcement learning algorithms.</li><li>• Collaborated with a team of 100+ people in implementing the first large-scale live demonstration of traffic intervention through automated vehicles.</li></ul>                                   |                                       |
|            | <b>Intern, Nissan Alliance SV Lab</b>                                                                                                                                                                                                                                                                                                                                                                                                                                           | <i>September 2021 - December 2021</i> |
|            | <ul style="list-style-type: none"><li>• Designed a framework for macroscopic traffic state estimation via connected vehicles.</li><li>• Implemented various traffic state estimation methods and validated them in simulation.</li><li>• Created visualization tools to monitor real-time variations in traffic.</li></ul>                                                                                                                                                      |                                       |
|            | <b>Intern, Toyota InfoTech Labs</b>                                                                                                                                                                                                                                                                                                                                                                                                                                             | <i>June 2021 - August 2021</i>        |
|            | <ul style="list-style-type: none"><li>• Built lightweight tools for simulating lane assignment strategies in open traffic.</li><li>• Developed a hierarchical mechanism for coordinating lane changing behaviors among connected vehicles.</li></ul>                                                                                                                                                                                                                            |                                       |
| TEACHING   | <b>Deep Multi-agent Reinforcement Learning with Applications to Autonomous Traffic</b>                                                                                                                                                                                                                                                                                                                                                                                          | <i>August 2018 - December 2018</i>    |
|            | <ul style="list-style-type: none"><li>• Developed the course curriculum and prepared homework problems in TensorFlow.</li><li>• Taught lectures on topics including non-stationary and communication in multiagent RL.</li></ul>                                                                                                                                                                                                                                                |                                       |
|            | <b>Introduction to Computer Programming for Scientists and Engineers</b>                                                                                                                                                                                                                                                                                                                                                                                                        | <i>January 2017 - May 2017</i>        |
|            | <ul style="list-style-type: none"><li>• Led lab sessions and mentored students through their early development.</li><li>• Formulated homework problems and exams in Matlab.</li></ul>                                                                                                                                                                                                                                                                                           |                                       |